Bayesian inference in IV regressions

Giannone, Lenza, Primiceri (ECB Working Paper Series)

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Causal inference with an endogenous regressor

Canonical IV model

$$y = x\beta + \delta\nu + \varepsilon, \qquad x = Z\pi + \nu.$$

- β: causal effect of x on y.
- ν: endogenous component of x.
- Z: exogenous instruments.
- π: first-stage coefficients.

The structural error is

$$u = \delta\nu + \varepsilon.$$

If $\delta \neq 0$, then

$$\mathrm{Cov}(x, u) = \delta \ \mathrm{Var}(\nu) \neq 0,$$

so OLS is inconsistent.

IV isolates the exogenous component of x:

$$\underbrace{x}_{\text{observed}} = \underbrace{Z\pi}_{\text{instrument-generated}} + \underbrace{\nu}_{\text{endogenous}}.$$

IV solves endogeneity only insofar as Z generates informative variation in x.

Weak instruments: identification becomes weak

First stage and instrument strength

$$x = Z\pi + \nu, \qquad \mu^2 = \frac{\pi' Z' Z\pi}{\sigma_\nu^2}.$$

μ^2 is the aggregate signal-to-noise ratio of the instruments.

$$\mathrm{Var}(\beta \mid \cdot) \approx \frac{\sigma_\varepsilon^2}{\sigma_\nu^2}\frac{1}{\mu^2}.$$

Precision about β

μ^2	Information about β
100	Precise; variance multiplier 0.01
0.1	Very limited; multiplier 10
0	β is not separately identified

Identification

When μ^2 is near zero, the data cannot distinguish the causal effect β from the endogeneity channel $\delta\nu$.

Standard TSLS can be confidently wrong

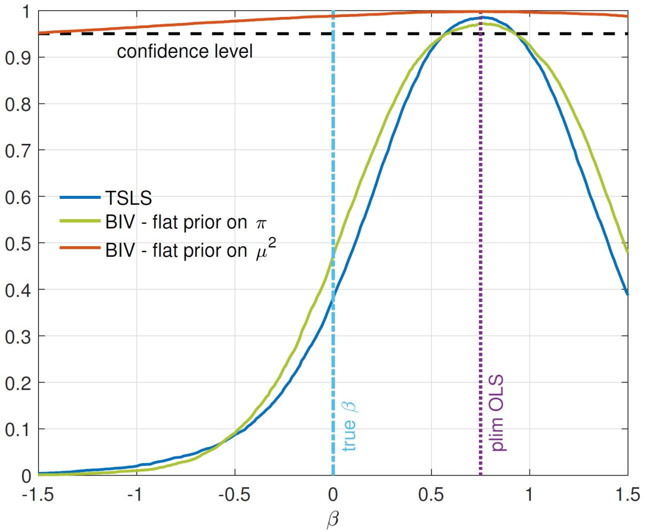

Figure 1: frequency with which each candidate value lies in a nominal 95% confidence interval. 1. Estimate the model 10000 times. 2. For every β, how often it was in the 95% credible interval.

Monte Carlo design

$$T = 250, \quad k = 10, \quad \beta = 0,$$
$$\delta = 0.75, \quad \pi = 0.$$

The instruments are completely irrelevant, so β is not separately identified.

TSLS result

- The true $\beta = 0$ is covered only about **40%** of the time.
- Values concentrate near the biased probability limit of the OLS estimator:

$$\beta + \delta = 0.75$$

- Standard errors of the TSLS are too tight.

Bayesian IV: why the same problem appears

$$p(\theta \mid \text{data}) \propto \underbrace{L(\text{data} \mid \theta)}_{\text{what the data say}} \quad \underbrace{p(\theta)}_{\text{prior weights}} \quad , \qquad \theta = (\beta, \delta, \pi, \dots).$$

The naive prior

Assign the same density to every vector π.

$$p(\pi) \propto 1.$$

The posterior is still overconfident

With irrelevant instruments, naive Bayesian IV (NB-IV) behaves almost like TSLS in Figure 1:

- it concentrates near the OLS probability limit;
- its credible intervals are far too narrow;
- it fails to represent non-identification.

The core mechanism: flat in π is not flat in instrument strength

Normalize $Z'Z = I$ and $\sigma_\nu^2 = 1$. Then

$$\mu^2 = \pi'\pi = \|\pi\|^2, \qquad \text{and the distance from the origin is} \qquad r = \|\pi\|.$$

Geometry of a flat density in k dimensions

The volume of a thin shell (all points whose distance from the origins lies between r and $r + dr$) is:

$$\text{(shell volume) } dV \propto r^{k-1}\, dr.$$

Probability mass is:

$$dP \propto p(\pi) dV \quad \text{so with} \quad p(\pi) \propto 1, \quad dP \propto r^{k-1}\, dr$$

Because $\mu^2 = \|\pi\|^2 = r^2$,

$$dr = \frac{d\mu^2}{2r} \implies \boxed{p(\mu^2) \propto (\mu^2)^{(k-2)/2}}.$$

Geometry derivation

The induced prior depends on k

k	$p(\mu^2)$	Implication
1	$(\mu^2)^{-1/2}$	favors values near zero
2	1	genuinely flat
10	$(\mu^2)^4$	strongly favors large strength

As k grows, shell volume rises, so flat $p(\pi)$ favors large μ^2.

Why the posterior becomes too precise

Flat prior on π

$\Downarrow$

High prior weight on large μ^2

$\Downarrow$

$\mathrm{Var}(\beta \mid \cdot) \propto 1/\mu^2$

$\Downarrow$

Artificially precise posterior for β

Proposition 1: make the prior flat where it matters

Objective

Place neutral prior weight directly on instrument strength:

$$\boxed{p(\mu^2) \propto 1.}$$

The geometry of π contributes

$$(\mu^2)^{(k-2)/2}.$$

Cancel it with the radial factor

$$(\mu^2)^{-(k-2)/2}.$$

Paper's weak-instrument-robust prior

$$\boxed{p(\pi \mid \sigma_\nu^2) \propto \left(\frac{\pi' Z' Z \pi}{\sigma_\nu^2}\right)^{-\frac{k-2}{2}} \left(\frac{1}{\sigma_\nu^2}\right)^{\frac{k}{2}}}$$

Therefore,

$$p(\mu^2) \propto (\mu^2)^{-\frac{k-2}{2}} (\mu^2)^{\frac{k-2}{2}} = 1.$$

Interpretation. For $k > 2$, the prior shrinks first-stage coefficients toward zero. That way the high-dimensional effect is offset and the prior on instrument strength μ^2 is flat.

Source: Giannone, Lenza, and Primiceri, Proposition 1.

Does the correction actually work?

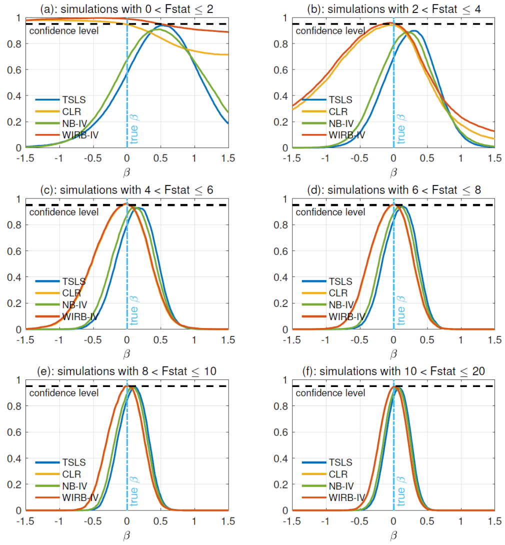

Figure 2: Frequency of inclusion in the 95-percent confidence interval.

Four methods

- TSLS
- NB-IV: flat prior on π
- WIRB-IV: flat prior on μ^2 **(This paper)**
- CLR: weak-IV-robust benchmark (Moreira 2003)

What Figure 2 shows

- At very low F, TSLS and NB-IV are badly distorted.
- WIRB-IV reports much greater uncertainty when identification is weakest.
- For $2 < F \leq 4$, WIRB-IV is already close to CLR.
- Above $F = 4$, WIRB-IV and CLR are nearly indistinguishable.

Angrist–Krueger: a real weak-IV laboratory

Returns to schooling

$$y = \log(\text{wage}), \qquad x = \text{years of schooling}.$$

Ability can raise both schooling and wages, making schooling endogenous.

Instrument and first stage

(Bound et al. 1995) Quarter of birth, interacted with year and state of birth, exploits compulsory-schooling laws.

$$n = 329509, \qquad k = 177, \qquad F = 2.43.$$

The first stage is very weak.

Paper's Table 1

Method	Actual estimate	Artificial data
TSLS	0.083 $[0.065, 0.102]$	98.2%
NB-IV	0.083 $[0.064, 0.101]$	97.6%
WIRB-IV	**0.100** $[0.070, 0.132]$	**0.2%**
CLR	NA (only yields confidence sets) $[0.065, 0.127]$	4.4%

Table 1: "Actual data" reports point estimates and 95-percent intervals. The column "Artificial Data" displays the percentage of simulations with 500 fake sets of instruments yielding statistically significant estimates at the 5-percent level. Replace quarter of birth with random quarters and interact them with year and state of birth. Instruments are random so they should contain no information about schooling.

TSLS and naive Bayes find effects with almost every random instrument set; WIRB-IV largely stops doing so.

Source: Giannone, Lenza, and Primiceri, Table 1; actual first-stage $F = 2.43$.

Takeaway

Problem

Weak instruments

$\Downarrow$

weak identification of β

Mechanism

Flat in π

$\neq$

uninformative about μ^2

Correction

Flat in μ^2

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uncertainty reflects identification

The paper's contribution

- Standard diffuse priors concentrate mass on regions of the parameter space where the instruments are strong.
- The authors construct a corrected prior that removes this distortion.
- The resulting Bayesian credible sets are asymptotically equivalent to Moreira's frequentist confidence sets.
- The correction prevents the prior from creating identification.

Appendix: geometry behind the induced prior

Full change-of-variables derivation

$$\mu^2 = r^2 \implies d(\mu^2) = d(r^2) = 2r\,dr \implies \boxed{dr = \frac{d\mu^2}{2r}} \qquad \text{(ordinary change of variables)},$$

$$\begin{aligned} dP \propto r^{k-1} dr &\implies dP \propto r^{k-1}\frac{d\mu^2}{2r} = \frac{1}{2} r^{k-2} d\mu^2, \\ r^2 = \mu^2 &\implies r^{k-2} = (r^2)^{(k-2)/2} = (\mu^2)^{(k-2)/2}, \\ &\implies dP \propto (\mu^2)^{(k-2)/2} d\mu^2. \end{aligned}$$

Read off the density

$$dP = p(\mu^2) d\mu^2 \quad \implies \quad \boxed{p(\mu^2) \propto (\mu^2)^{(k-2)/2}}.$$

The density $p(\mu^2)$ is the term multiplying the infinitesimal interval $d\mu^2$ in the probability expression.

$$\boxed{dP \propto r^{k-1} dr} \implies \boxed{dr = d\mu^2/(2r)} \implies \boxed{dP \propto (\mu^2)^{(k-2)/2} d\mu^2} \implies \boxed{p(\mu^2) \propto (\mu^2)^{(k-2)/2}}$$

Back to Proposition 1